Networking Academy

CCNP Enterprise: Core Networking (ENCOR)

Scope and Sequence

Version v9

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Target Audience

The Cisco Networking Academy® CCNP Enterprise curriculum is designed for participants who are seeking professional-level jobs in the ICT industry or hope to fulfill prerequisites to pursue other CCNP or CCIE certifications. The entire curriculum is appropriate for learners at universities and colleges.

Prerequisites

CCNA or equivalent knowledge and skills.

Course Description

In this curriculum, Cisco Networking Academy™ participants learn, apply, and practice CCNP Enterprise knowledge and skills through a series of in-depth, hands-on experiences that reinforce their learning.

The CCNP Enterprise v9 curriculum is presented in two courses that provide integrated and comprehensive coverage of professional-level networking technologies.

Upon completion of each course, learners will be prepared to take the certification exam associated with that course.

The following two courses comprise the CCNP Enterprise v9 curriculum:

CCNP Enterprise: Core Networking v9

The Implementing and Operating Cisco Enterprise Network Core Technologies (ENCOR) course provides learners with the knowledge and skills needed to install, configure, operate, and troubleshoot an enterprise network. This course introduces overlay network design through SD-Access and SD-WAN solutions and teaches learners to implement security principles and automation and programmability within an enterprise network. This course helps prepare learners to take the 350-401 ENCOR exam, which is part of the CCIE Enterprise Infrastructure certification. This version is aligned with the updated version of the 350-401 exam, version 1.2.

Learners will be able to:

- Configure and implement identified solutions by applying planned implementation processes using Cisco IOS Software commands and applications.
- Verify appropriate show and debug commands and applications to ensure correct solution implementation and performance.
- Troubleshoot appropriate show and debug commands and applications to identify the cause of basic-level network issues and correctly implement a solution that ensures the network is performing as desired.

CCNP Enterprise: Advanced Routing v9

The Implementing Cisco Enterprise Advanced Routing and Services (ENARSI) course provides learners with the knowledge and skills needed to install, operate, manage, and troubleshoot an enterprise network. Throughout the course, learners will build the skills necessary to ensure the smooth and effective operation of these networks. Completing this course prepares them to take the Implementing Cisco Enterprise Advanced Routing and Services (300-410 ENARSI)

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exam version 1.1. Learners who pass the exam will earn the Cisco Certified Specialist - Enterprise Advanced Infrastructure Implementation certification, which is also part of the CCNP Enterprise certification.

Learners will be able to:

- Install, configure, and operate enterprise networks.
- Troubleshoot enterprise networks with a focus on advanced routing and services.
- Implement Layer 3 technologies, route manipulation, VPN technologies, network infrastructure, and network security solutions.
- Demonstrate advanced knowledge of enterprise networking tools, technologies, and concepts.

These two courses, along with the corresponding certification exams, align to the overall CCNP Enterprise certification. Both courses provide learners with extensive opportunities for hands-on practical experience and career-skills development.

Authorized Use

Access to the CCNP curriculum is granted exclusively under the terms of the Cisco Networking Academy Membership Agreement.

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Course Features

The CCNP Enterprise v9 curriculum includes many features to engage and assess learner knowledge and skills.

Curriculum

The online curriculum has been repurposed for use by the Cisco Networking Academy from the premium offerings at Cisco U.

The Cisco U. built-in, hands-on labs, (using the terminology “Discovery”), available to Cisco U. premium subscribers, have been stripped and are only available at Cisco Networking Academy as video-based demonstrations.

The course at Cisco Networking Academy has been expanded with additional hands-on practice activities that leverage the Cisco Hardware Classroom Equipment, Cisco Packet Tracer, or Cisco Modeling Labs free edition (CML-Free). These lab activities are threaded throughout the course using the Assignment Flow.

Assessments

Assessments and practice activities are focused on specific certification competencies to increase knowledge and skill retention.

Embedded module-group assessments provide immediate feedback to support the evaluation of knowledge and acquired skills.

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Activities

Hands-on labs help students develop critical thinking and complex, problem-solving skills.

It is recommended to leverage CML-Free to emulate real network operating systems and practice important network virtualization skills and concepts, such as a digital twin.

Cisco Packet Tracer simulation-based learning activities are added as optional instructor resources to support vital CCNA skills.

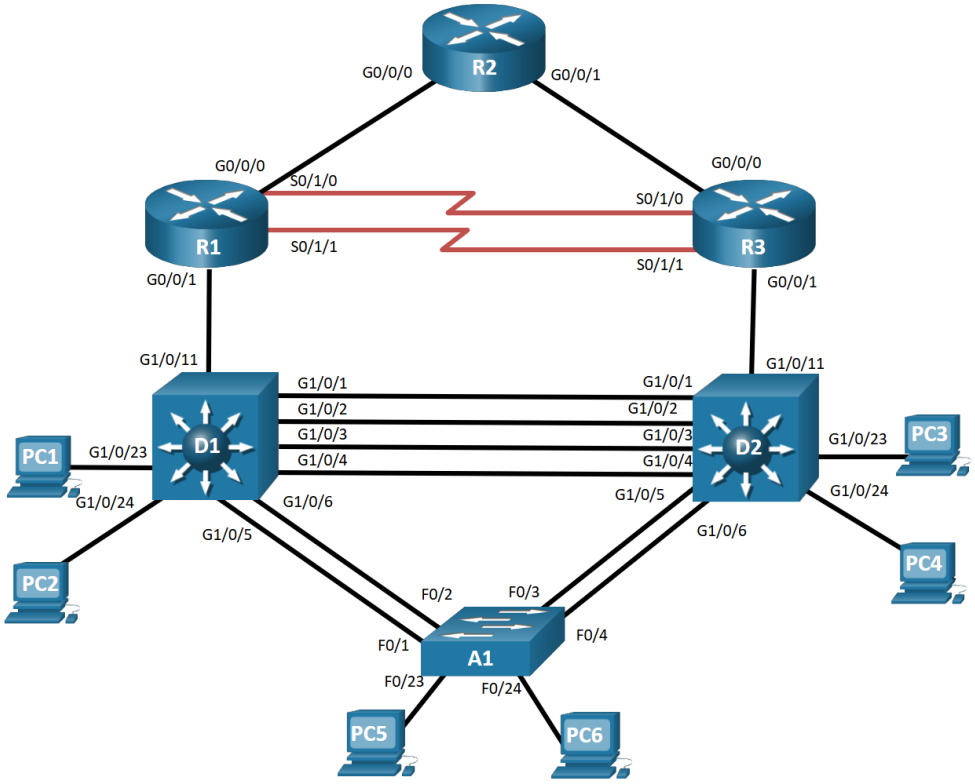
Video demonstrations give insight into complex workflows and processes providing exposure to advanced networking tools and operations.

Equipment Requirements

Detailed equipment information, including descriptions and part numbers, is available in the CCNP Enterprise Equipment List, which is located on the Cisco NetAcad Equipment Information site (https://www.netacad.com/resources/program-info/equipment-information).

The labs in the CCNP Enterprise curriculum use the following topology and equipment.

CCNP Enterprise Baseline Physical Topology (for both ENCOR and ENARSI)



Baseline Equipment Bundle

- 3x C8200L-1N-4T (2 with NIM-2T)
- 2x C9300L-24T-4G-A
- 1x C9200L-24T-4G-E
- Ethernet cables as shown in the topology
- 2x CAB-SS-V35MT= (10' DTE Serial Cable)

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- 2x CAB-SS-V35FC= (10' DCE Serial Cable)
- PCs - minimum system requirements
    - CPU: Intel Core i5 (8th Gen or newer) / AMD Ryzen 5 or equivalent
        - Must support hardware virtualization (Intel VT-x or AMD-V) for lab simulation software
    - Operating System, such as Microsoft Windows, Linux, or Mac OS
    - RAM: 16 GB (Recommended for running CML/Virtual Machines); 8 GB (Minimum)
    - Storage: 20 GB of free disk space
    - Ethernet port (RJ-45) or USB Ethernet adapter (for connecting to physical lab equipment)
    - Display resolution: 1920 x 1080 (FHD) or higher
    - Latest video card drivers, operating system updates, and a modern web browser (Chrome, Firefox, or Edge)
- Internet connection for lab and study PCs

Software

- Cisco IOS versions:
    - Routers: Version IOSXE-17.15.4c or higher.
    - Layer 3 Switches: Version IOS-XE 17.12.6 or higher.
    - Layer 2 Switches: Version IOS 17.12.6 or higher
- Packet Tracer v9 or newer
- Open-source server software for various services and protocols, such as HTTP, DHCP, FTP, TFTP, etc.
- Terminal emulation and SSH client software, such as Tera Term and PuTTY for lab PCs
- Oracle VirtualBox, latest version
- VMware Workstation, latest version
- Wireshark version: latest stable version
- Terminal emulation software for the installed PC operating system

Course Outline

The CCNP Enterprise: Core Networking course is structured into 6 domains (module groups) that cover specific competencies. The table below details the modules and their associated objectives. Each module is an integrated unit of learning that consists of content, activities, and assessments that target a specific set of competencies. The size of the module depends on the depth of knowledge and skill needed to master the competencies.

No.	Module Name	Objective
Network Switching		
1	Examining Cisco Enterprise Network Architecture	Examine the Cisco Enterprise Architecture models (Campus, Edge, SP Edge, Remote) and the layers of the Enterprise Campus.
2	Cisco Switching Paths	Analyze Layer 2 switch operations, the role of the CAM table, switching mechanisms, and Cisco Express Forwarding.
3	Implementing Campus LAN Connectivity	Implement and troubleshoot VLANs, trunks, and inter-VLAN routing.
4	Building Redundant Switched Topology	Configure and fine-tune Spanning Tree Protocol (STP) to resolve issues in redundant switched topologies.

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No.	Module Name	Objective
5	Implementing Network Redundancy	Implement First Hop Redundancy Protocols (FHRP) to ensure network availability.
6	Implementing Layer 2 Port Aggregation	Implement EtherChannel for bandwidth aggregation, load balancing, and network troubleshooting.
7	Introducing QoS	Explain application network requirements (bandwidth, jitter, latency) and implement Quality of Service (QoS) mechanisms.
Network Routing		
8	Explaining EIGRP	Explain the operation, benefits, and characteristics of EIGRP compared to OSPF.
9	Implementing OSPF	Implement OSPF neighbor adjacencies, LSDB usage, LSA types, and configure single and multi-area OSPF.
10	Optimizing OSPF	Optimize OSPF networks using cost tuning, route summarization, filtering, and OSPFv3 configurations.
11	Exploring EBGP	Explore the fundamentals and implementation of the Border Gateway Protocol (BGP).
12	Introducing Multicast Protocols	Explain the benefits and fundamental concepts of multicast technology.
13	Implementing NAT	Implement various NAT address types, configurations, and the NAT Virtual Interface (NVI).
Network Security		
14	Implementing Infrastructure Security	Implement Access Control Lists (ACLs) and control plane policing to permit, deny, or rate-limit traffic.
15	Exploring Enterprise Network Security Architecture	Explore enterprise network security architecture, potential threats, and security tools for protection.
16	Implementing Secure Access Control	Implement AAA services using TACACS+ and RADIUS servers for secure access control.
Virtualization and Network Services		
17	Introducing Virtualization Protocols and Techniques	Introduce virtualization protocols and techniques for virtualizing devices and network paths.
18	Exploring Virtual Private Networks and Interfaces	Explore the purpose, benefits, and deployment options of Virtual Private Networks (VPNs).
19	Implementing Network Services	Implement network services for information collection, event notification, and troubleshooting support.

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No.	Module Name	Objective
20	Using Network Analysis Tools	Apply troubleshooting concepts, diagnostics procedures, and network analysis tools.
Programmability		
21	Discovering the Basics of Python Programming	Discover the basics of Python programming and its benefits for network automation over traditional CLI.
22	Introducing Network Programmability Protocols	Introduce network programmability protocols and automation tools for efficient network operations.
Software-Defined Networking		
23	Exploring Cisco DNA Center-Network Automation and Management	Explore Cisco Catalyst Center (DNA Center) for implementing business intent, automation, and assurance.
24	Examining the Cisco SD-Access Solution	Examine the benefits and components of the Cisco SD-Access solution.
25	Exploring the Working Principles of the Cisco SD-WAN Solution	Explore the working principles, benefits, and components of the Cisco Catalyst SD-WAN solution.
26	Introducing APIs in Cisco DNA Center and vManage	Introduce Southbound and Northbound APIs (REST, Intent) used in Cisco Catalyst Center and SD-WAN Manager.

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